

How we faked it

A dependent function $(p : P) \rightarrow Rp$, in a language that has none

September 1, 2026

Abstract

Ghani’s containers turn on a dependent function: the type of the reply depends on the prompt. TypeScript has no dependent types. This note is about the one substitution that makes the whole edifice stand — a generic method whose return type is a conditional type applied to the inferred type argument — what it genuinely buys, exactly where it degrades, and the one place the project abandons it altogether. Along the way it disposes of a natural suspicion: that `async` had something to do with it. It did not.

1 The thing to be faked

An inhabitant of a container (P, R) is a dependent function

$$(p : P) \rightarrow Rp$$

in which the *type* of the result is computed from the *value* of the argument. Ask for a harvest and you get a harvest report; ask for a census and you get a census return; and no type in the language relates the two except through p itself.

TypeScript has no such construct. What it has is: generic functions, which infer a type argument from a value argument; and conditional types, which compute a type from a type. Put together they give this:

```
run<S extends Shape<C>>(shape: S): PosAt<C, S>
```

Three ingredients, and it is worth being precise about which does what.

generic inference	S is recovered from the argument — this is the only channel by which anything about p reaches the type level
conditional type	$\text{PosAt}\langle C, S \rangle$ computes the position type from S , by distributing over the union of fibres
a finite index	makes that distribution <i>exact</i> : over a finite S , $\Sigma(s : S). B s = \bigcup_s \{s\} \times B s$

2 Three things that are not the same thing

It is easy — and this note managed it for several drafts — to run together three decisions that are in fact independent:

	what it is	what it needs
the algebra	prompts, positions, the four combinators	generics, conditional types, a finite index. Nothing else.
the location	where the handler lives: this process, another process, another machine	a function call, a spawn, or HTTP
the blocking	whether the caller waits	independent of both above

HTTP does not force asynchrony in your own code; asynchrony does not require a network (`setTimeout` touches nothing); and the algebra requires neither. The rest of this section takes them apart one at a time.

async has nothing to do with it

The real signature carries a `Promise`, which invites the thought that concurrency is somehow load-bearing. It is not. Here is the same encoding with no `Promise`, no `await`, and no `socket`:

```
interface SyncRunner<C extends Cont> {
  readonly fibres?: C;
  run<S extends Shape<C>>(shape: S): PosAt<C, S>;    // no Promise
}
declare const agri: SyncRunner<Agri>;

const h = agri.run({tag: "Harvest", year: 1930});
const c = agri.run({tag: "Census"});
const grain: number = h.grain;           // typed HarvestReport, uncast
const hands: number = c.workforce;       // typed CensusReturn, uncast
// @ts-expect-error a Census reply has no 'grain'
const wrong: number = c.grain;
```

That file typechecks with zero errors, and the negative assertion fires — TypeScript reports an unused `@ts-expect-error` as an error in its own right, so its silence is evidence that the dependency really bites.

So why is the real thing async? Because the reply arrives from another process over a socket. Causation runs the other way from the suspicion: we chose real process boundaries, and *that* forced `Promise`. Ghani’s own point in §3 is that what makes a program event-driven is the prompt-indexed response type, not the event loop — strip the waiting away and the container is what remains.

One thing async does buy. It makes $\boxtimes$ against $\triangleleft$ visible in the *execution* and not only in the types. The tensor fires both prompts at once and requires both replies; the sequential composition cannot even construct its second prompt until the first reply is in hand:

```
const [l, r] = await Promise.all([a.run(s.left, d), b.run(s.right, d)]); // (x)
const first  = await a.run(s.first, d);
const second = await b.run(s.next(first), d);                          // <|
```

Synchronously these would be two calls in a row either way, and the difference would live entirely in the types.

A correction. The source comment on that `Promise.all` says it “is not an optimisation here; it is the definition”. That is an overstatement, and writing a synchronous tower disproves it. The tensor is defined by both shapes being supplied *up front* and both replies being owed — and that independence is enforced by the shape type $\{\text{left} : S_A, \text{right} : S_B\}$, not by concurrency. You cannot make the right prompt depend on the left reply, because you had to construct it before calling `run`. `Promise.all` adds parallelism, not meaning.

So it could have been synchronous — and here it is

`sync-tower.ts` is the same construction with no `async`, `await` or `Promise` anywhere in the algebra. A hop blocks:

```
new Deno.Command("deno", {args: [...delegate...], stdout: "piped"}).outputSync()
```

It runs a tensor over two live commissariats and returns typed, uncast positions. This inverts how the architecture looks. Deno has no synchronous `fetch` and no synchronous HTTP client at all, so an *in-process* tower would have had no choice but to be `async`. It is precisely because a hop is a spawned CLI tool that the algebra can block: the child does the asynchronous fetch inside its own event loop, and the parent simply waits. The decision that looks more distributed is the one that made synchrony available.

What cannot be synchronous is the *listening* side — `Deno.serve` handlers are `async` and there is no blocking HTTP server. But that is outside the algebra, and the line falls exactly

where §3 says it should: **the container algebra can be synchronous; the event loop cannot.**

And it needed no wire at all

Strip the last of it away. `local-tower.ts` is the same algebra with no HTTP, no processes, no event loop and no `async` — a leaf is a table of handler functions, and a hop is a call:

```
function local<C extends Cont>(label: string, handlers: ...): Runner<C> {
  return { label, run: <S extends Shape<C>>(shape: S) =>
    handlers[(shape as {tag: string}).tag](shape as never) as PosAt<C, S> };
}

const t = tensor(fields, fields);
const pair = t.run(t.both({tag: "Harvest", hands: 105}, {tag: "Census"}));
const g: number = pair.left.grain;    // typed HarvestReport -- uncast
const h: number = pair.right.hands;   // typed CensusReturn  -- uncast

const s = seq(fields, rail);
const chain = s.run(s.step({tag: "Harvest", hands: 105},
  (r) => ({tag: "Haul", grain: r.grain}))); // <-- <|
```

It runs under `deno run` with *no permissions at all* — no `--allow-net`, no `--allow-run`, no `--allow-read` — because it touches nothing. The tensor and the sequential composition behave exactly as they do across the wire, and every position is typed and uncast.

So why is there a wire? Two reasons, neither of them about types. The first is that the brief asked for one: several servers, possibly on different machines, reached by a CLI tool. The second is the interesting one. Every commissariat holds a hidden efficiency and a bias — how much it inflates a shortfall — and the player plans against the reports rather than the truth. In a single process that privacy is a *convention*: module-scoped state, one stray import from collapsing. Across five processes with nothing but JSON between them it is enforced by the operating system, because there is no reference to reach through. The game's subject is an information asymmetry, and the wire is what makes the asymmetry a fact instead of a promise. It costs about a fifth of a second per hop.

3 Where the dependency is exact

The substitution works when the compiler can see *which fibre* a prompt lies in. Two cases, both verified:

```
const a = agri.run({tag: "Harvest", year: 1930});    // S = the literal shape
                                                    // -> {grain: number}  EXACT

const p: Shape<Agri> = {tag: "Harvest", year: 1930}; // declared as the union...
const b = agri.run(p);                               // ...but narrowed by the
                                                    // discriminant -> EXACT
```

The second was a surprise. A `const` declared at the union type but initialised with a literal is narrowed by its discriminant tag, so the dependency survives being written down as a variable. The encoding is more robust than it looks.

The introduction rules do the real work

Each combinator ships a constructor for its shapes, and each is generic in the fibre. That is what keeps `S` narrow through a composite. The sequential one is the deepest instance:

```
step<SA extends Shape<A>, SB extends Shape<B>>>(
  first: SA,
  next: (r: PosAt<A, SA>) => SB,
```

```
) : {first: SA; next: (r: PosAt<A, SA>) => SB}
```

SA is inferred from `first`, so `next` is contextually typed at *that fibre alone*: it receives the positions of the shape you actually gave, not a union over all of them. This is the Σ -type faked — not a pair of prompts, but a prompt together with a function from its replies to the next prompt. And it is why, in the game, `supplied.first.toPort` is a `HaulReport` field with no cast at the call site.

4 Where it degrades

The fake carries the dependency on the **static type** of the argument, never on its runtime value. So it fails precisely where the compiler cannot see a single fibre — and the commonest such place is a *forwarder*: any helper generic over the container’s prompts.

```
function forward(shape: Shape<Agri>) {
  return agri.run(shape);           // S is now the WHOLE union
}
const c = forward({tag: "Harvest", year: 1930});
// c : {grain: number} | {workforce: number}    <-- the dependency is gone
```

`PosAt` distributes over `S` as well as over `C`, so a union argument yields the union of *all* positions. Note what this is and is not. It is not unsound — you cannot read `grain` off it without narrowing first, and the compiler will stop you. But the dependency has been traded for a disjunction, and the caller must re-discover by hand what the type system knew one frame earlier.

This is the precise sense in which it is a fake. A genuine $(p : P) \rightarrow Rp$ returns Rp for the actual runtime p , always, including inside a forwarder. This returns the exact position only where the compiler statically knows the fibre, and the union everywhere else. The dependency is on the singleton type TypeScript infers, not on the value.

5 Where it is abandoned

At the outermost edge the project gives up on it entirely. Everything a human or a search program sends to the Plan goes through:

```
export async function ask(prompt: unknown): Promise<Dict>
```

`unknown` in, `Record<string, unknown>` out. Not a degraded dependency — no typing at all.

This is not laziness, and it is worth saying why. An interactive menu is a forwarder by construction: it holds a list of options the player might choose, it does not know which until a key is pressed, and a list of ten differently typed prompts is exactly the union case above. The same is true of the letter search, which replays arbitrary sequences. So the container discipline holds *inside* the tower, where Gosplan composes and calls its commissariats through `leaf` and the combinators, and stops at the boundary where prompts become data chosen at runtime.

Which is the honest shape of the result. Dependent typing survives exactly as far as the compiler can follow the control flow, and no further. The tower’s interior is typed; its front door is not.

6 The rest of the fakery, briefly

The container as its graph. R cannot be a function from values to types, so a container is presented as the union of its fibres, one `Fib` per prompt. `Fib` is never constructed — it is a type-level record, the object language of $\text{Fam}(\text{Set}^{\text{op}})$, taken apart only by `infer`.

The delegate leg as a CLI tool. $u : P \rightarrow Q$ is not a function call but a spawned process, `delegate.ts <port> <json> <depth>`, which knows nothing of any container: it carries

text. Control genuinely leaves the process. The file is located by `new URL("./delegate.ts", import.meta.url)`, resolved at run time and never checked — which is how a clean typecheck once produced a playthrough that died on its first hop.

The wire. JSON has six kinds of value and no notion of types, so a tag is a claim rather than a fact, and a wrong one usually does not crash: a missing property reads as `undefined`, so the handler returns HTTP 200 with a well-formed reply meaning nothing. Both directions are therefore validated rather than cast — 56 field checks across the five servers, and per-fibre decoders on the way back.

7 The bill

site	casts	what is asserted
<code>leaf</code>	2	the shape has a <code>tag</code> ; the decoded reply is the position
<code>sumC</code>	2	the shape is a tagged side; the reply is the position
<code>tensorC</code>	2	the shape is a pair; the pair of replies is the position
<code>productC</code>	2	the shape is a pair; <code>{side, value}</code> is the position
<code>seqC</code>	2	the shape is <code>{first, next}</code> ; <code>{first, second}</code> is the position
total	10	all inside <code>lib/algebra.ts</code> , 260 lines

Every `run` body must destructure a shape it knows only through a conditional type and re-assert the result; hence two per combinator. **None is in the game.** The game’s own casts are `as const` on literals and narrowing strings a validator has already checked; no call site in `stalin` ever asserts the type of a container position.

Two traps, both silent

The fibre test written backwards. `Extract<C, Fib<S, unknown>` keeps the fibres assignable to the query, but a shape you hold is usually *narrower* than its fibre’s declared shape. What is wanted is `S extends S1`. Written the other way, every position becomes `never` and nothing complains near the mistake.

Conditional types are not inference sites. If `C` occurs only inside `Shape<C>` and `PosAt<C, S>`, TypeScript cannot infer it: `seqC(a, b)` silently infers `A = Cont` and every downstream position becomes `unknown`. Hence the phantom `readonly fibres?: C`, a field that is never assigned and exists solely to give `C` an inferable position.

Summary

Exact	inline prompts, and <code>consts</code> narrowed by their discriminant: the position type follows from the prompt, uncast
Degraded	any forwarder generic over prompts: the union of all positions, sound but no longer dependent
Abandoned	the outermost client, <code>ask(unknown): Dict</code> — because a menu is a forwarder by construction
Cost	10 casts, one phantom field, and a finite index
Irrelevant	<code>async</code> , the wire, and the event loop — all three are about <i>where the handler lives</i> and <i>who waits</i> , not about the algebra. <code>local-tower.ts</code> runs it with none of them